



Introduction to NLP

AIAA 4051 Lecture 6

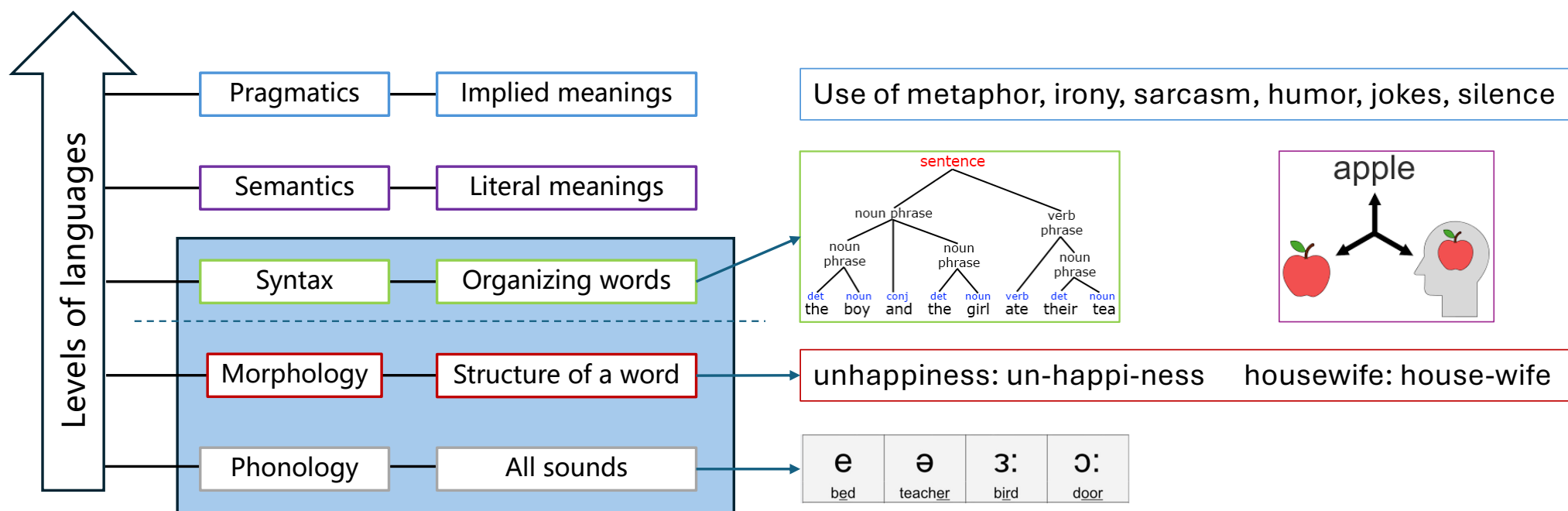
Instructor: Sihong Xie

Email: sihongxie@hkust-gz.edu.cn

AI Thrust, HKUST(GZ)

Grammar and syntax*

- There is a hierarchy of linguistic elements.
 - syntax is part of a grammar above word-level.



*Note: these are what NLP researchers focused on in the past before LLM, but now next token prediction+advanced architecture can skip such an intermediate processing step. We discuss these concepts/techniques to build intuition in NLP.

| Syntax

- Syntax are rules defining how words are organized into larger parts
 - Distinguish between legal and illegal sentences in a language
 - We would not say

** together words the way are arraged.*

since it is not how words are organized in English.

- Rather, we would say

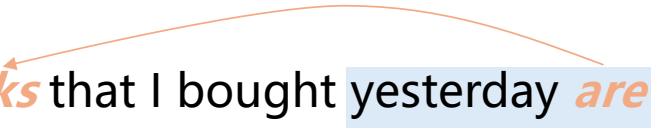
the way words are arraged together.

- Legal syntax makes communication easier.
- Fluency: how familiar a speaker/writer can use the syntatic rules.
 - A baby learn syntax through using a language, without knowing the rules.
 - Second language learners learn the rules explicitly.

| n-grams and HMM are modeling shallow syntax

- Shallow syntax: local structure of a language.
 - *the* followed by a noun
 - Subject-Verb agreement: *he drinks* vs. *they drink*
 - etc.
- n-gram and HMM are Markovian
 - Use near history (usually the immediate preceedor) to predict current word.
 - Cannot handle long-range dependencies
- Deep syntax: global and long-range dependencies

The *books* that I bought yesterday *are* expensive.

A curved orange arrow originates from the word 'books' and points to the word 'are', illustrating a long-range dependency that cannot be handled by n-gram or HMM models.

- *are* depends on *books* and is in its plural form.
- but distance between *books* and *are* is too large for n-gram and HMM to handle.

Constitute

- Constitute: a group of words that acts as a single grammatical unit.

- Example units

Harry the Horse
the Broadway coppers
they

a high-class spot such as Mindy's
the reason he comes into the Hot Box
three parties from Brooklyn

- Use of the units

three parties from Brooklyn *arrive...*
a high-class spot such as Mindy's *attracts...*
the Broadway coppers *love...*
they *sit*

- a **unit** can appear in different places in a sentence without changing its semantics

On September seventeenth, I'd like to fly from Atlanta to Denver

I'd like to fly on September seventeenth from Atlanta to Denver

I'd like to fly from Atlanta to Denver on September seventeenth

- a **unit** can't be broken down while maintaining its semantics.

*On September, I'd like to fly seventeenth from Atlanta to Denver

*On I'd like to fly September seventeenth from Atlanta to Denver

*I'd like to fly on September from Atlanta to Denver seventeenth

| Context-free grammar

- Context-free grammar is a mathematical system modeling *constituent* structures.

- Context: the other elements in a sentence outside the constitute

On September seventeenth,	I'd like to fly from Atlanta to Denver
I'd like to fly	on September seventeenth from Atlanta to Denver
I'd like to fly from Atlanta to Denver	on September seventeenth

- Context-free: the constitute's role and semantics won't change with respect to the context
 - In the above examples, "On September seventeenth" means the same and is modifying the action "fly", regardless of the location of the constitute.

Context-free grammar

- A CFG consists of 4 elements:

N	a set of non-terminal symbols (or variables)
Σ	a set of terminal symbols (disjoint from N)
R	a set of rules or productions, each of the form $A \rightarrow \beta$, where A is a non-terminal, β is a string of symbols from the infinite set of strings $(\Sigma \cup N)^*$
S	a designated start symbol and a member of N

- Derivations: if $A \rightarrow \beta$ is a production of R and α and γ are any strings in the set $(\Sigma \cup N)^*$, then we say that $\alpha A \gamma$ **directly derives** $\alpha \beta \gamma$, or $\alpha A \gamma \Rightarrow \alpha \beta \gamma$.

$(\Sigma \cup N)^*$ is the set of strings of symbols from $\Sigma \cup N$ of any length.

○ m steps of derivation from α_1 to α_m $\alpha_1 \Rightarrow \alpha_2, \alpha_2 \Rightarrow \alpha_3, \dots, \alpha_{m-1} \Rightarrow \alpha_m$

- Language derived from CFG: $\mathcal{L}_G = \{w \mid w \text{ is in } \Sigma^* \text{ and } S \xRightarrow{*} w\}$

Context-free grammar

• Example

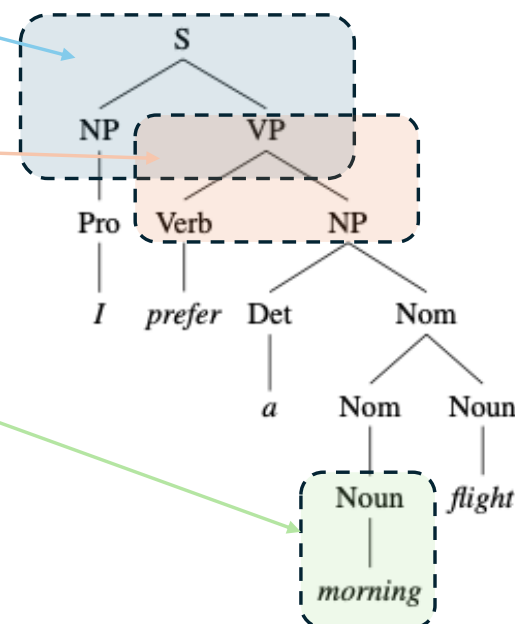
Rules generating non-terminals

Grammar Rules	Examples
$S \rightarrow NP VP$	I + want a morning flight
$NP \rightarrow$ <i>Pronoun</i>	I
<i>Proper-Noun</i>	Los Angeles
<i>Det Nominal</i>	a + flight
$Nominal \rightarrow$ <i>Nominal Noun</i>	morning + flight
<i>Noun</i>	flights
$VP \rightarrow$ <i>Verb</i>	do
<i>Verb NP</i>	want + a flight
<i>Verb NP PP</i>	leave + Boston + in the morning
<i>Verb PP</i>	leaving + on Thursday
$PP \rightarrow$ <i>Preposition NP</i>	from + Los Angeles

Rules generating terminals, and this is just deterministic POS-tagging.

<i>Noun</i>	$\rightarrow$ flights breeze trip morning
<i>Verb</i>	$\rightarrow$ is prefer like need want fly
<i>Adjective</i>	$\rightarrow$ cheapest non-stop first latest other direct
<i>Pronoun</i>	$\rightarrow$ me I you it
<i>Proper-Noun</i>	$\rightarrow$ Alaska Baltimore Los Angeles Chicago United American
<i>Determiner</i>	$\rightarrow$ the a an this these that
<i>Preposition</i>	$\rightarrow$ from to on near
<i>Conjunction</i>	$\rightarrow$ and or but

A parsing tree representing multiple steps of derivations using the CFG to generate a sentence.



| Context-free grammar: sentences

- Declarative
 - $S \rightarrow NP VP$ (*I want a flight from Ontario to Chicago*)
- Imperative
 - $S \rightarrow VP$ (*Show the lowest fare*)
- Yes-no questions
 - $S \rightarrow Aux NP VP$ (*Do any of these flights have stops?*)
- Wh-structures
 - $S \rightarrow Wh-NP VP$ (*What airlines fly from Burbank to Denver?*)
 - $S \rightarrow Wh-NP Aux NP VP PP$ (*What flights do you have from Burbank to Denver?*)

Context-free grammar: noun phrases (NP)

- $NP \rightarrow (Det) (Card) (Ord) (Quant) (AP) \text{ Nominal}$
 - () means the enclosed non-terminal is optional
 - $Card \rightarrow one, two, \dots$; $Ord \rightarrow first, second, \dots$; $Quant \rightarrow many, (a) few, several$
 - $AP \rightarrow (RB) JJ$: *least expensive*
- Determiners (Det)
 - Simple lexical determiners: a, an, the
 - Possessive ($Det \rightarrow NP's$): *United's flight*
 - Together with $NP \rightarrow Det NP$
 - $Det \Rightarrow NP's \Rightarrow Det NP's \Rightarrow NP's NP's \Rightarrow Det NP's NP's \Rightarrow NP's NP's NP's$
 - *Denver's mayor's mother's canceled flight*
 - From the above example, we can see the derivation can be recursive.

| Context-free grammar: nominals

- Nominals: similar to noun but in more forms to gain more information
 - simple nominals: Nominal→Noun
 - this serves as the bottom of a parsing tree with no more recursion.
 - more complex nominals:
 - Nominal→ Nominal Noun (*morning flight*)
 - Nominal→ Nominal PP (*flight to Boston*)
 - Nominal→ Nominal Gerundive-VP (*flight leaving before 10*) More on VP later.
 - Nominal→ Nominal ed-VP (*dinner served on the flight*)
 - Nominal→ Nominal infinitive (*flight to arrive in Boston*)
 - Nominal →relative-clause (*flight that serves breakfast*)

| Context-free grammar: verb phrases (VP)

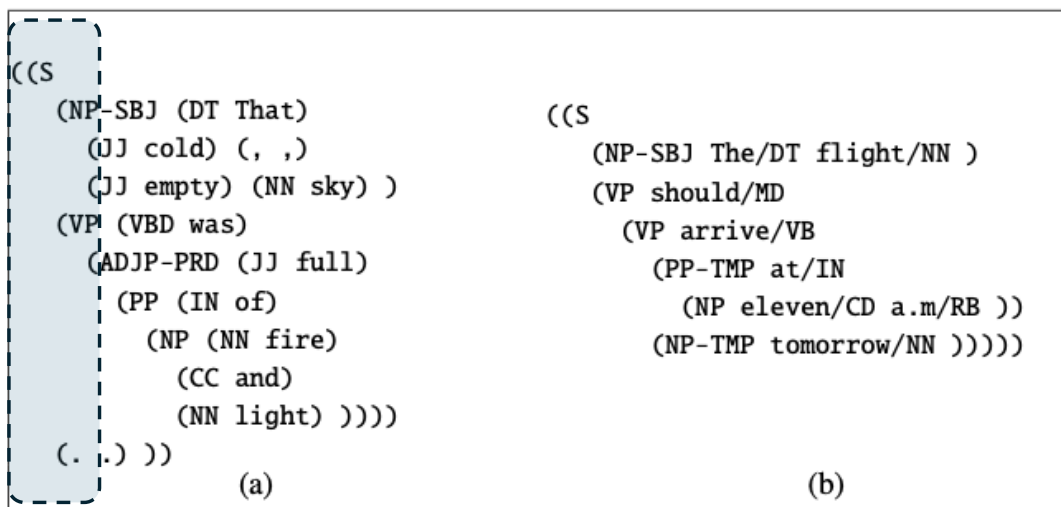
Examples of VP

- VP→Verb (*disappear*)
- VP→Verb NP (*prefer a morning flight*)
- VP→Verb NP PP (*leave Boston in the morning*)
- VP→Verb PP (*leaving on Thursday*)
- VP→Verb VP (*want to fly from Milwaukee to Orlando*)
- VP→Verb S (You said you had a two hundred sixty six dollar fare)
 - S is called “sentential complement”

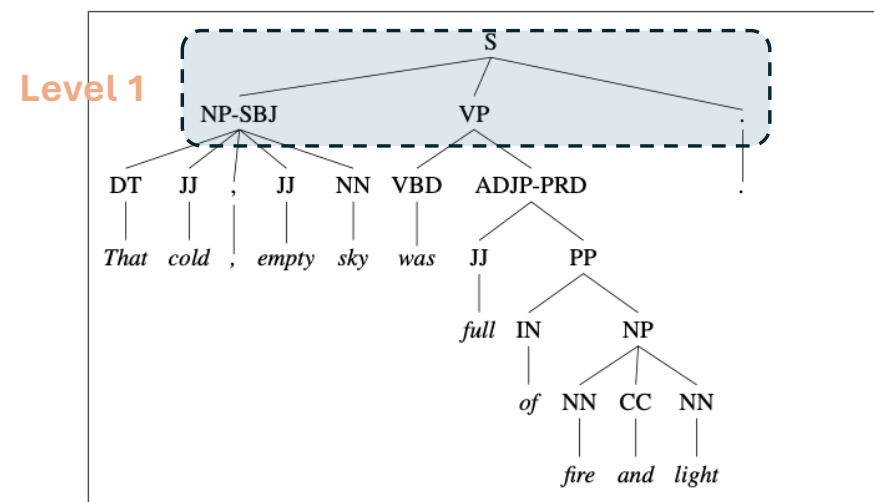
Where to obtain a CFG?

- Penn Treebank: a corpus with sentences annotated with parsing trees.
 - derive CFG grammars from the data;
 - can serve as training data for syntactic parsers.

Level 1 Nested format



Tree format



The bitter lesson

- To spend time on building human knowledge in models, or on building scalable models to discover knowledge from massive data?

Right figure shows algorithms over time beat human players.

Green: algorithms that have built-in human knowledge.

Red and Purple: data-driven methods that learn to play by themselves.

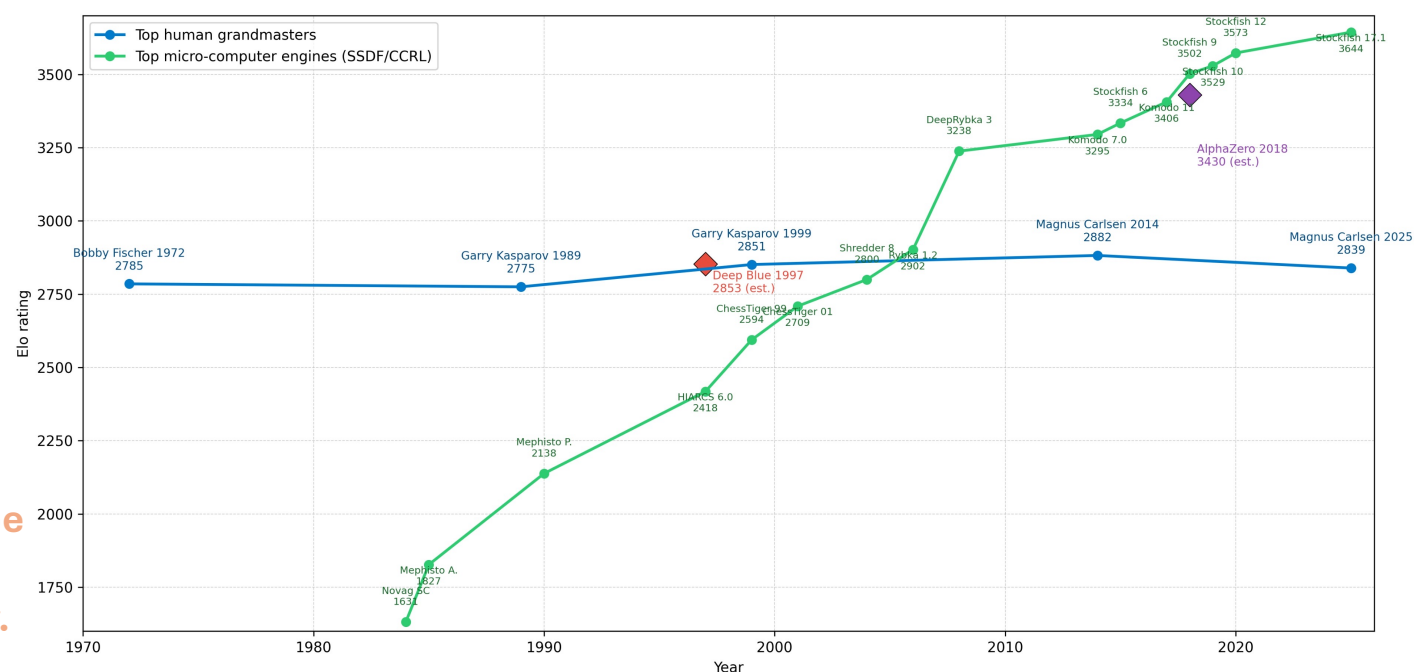
Grammar is human knowledge to be built into an NLP model.

With LLM, this seems unnecessary.

Original article: <http://www.incompleteideas.net/IncIdeas/BitterLesson.html>

Image source: https://www.dbreunig.com/img/chatgpt_agent_chess.jpg

Elo Ratings of Top Human Grandmasters vs Top Chess Engines (1972-2025)



The green line tracks the strongest commercially available micro-computer engines (SSDF/CCRL). Deep Blue (IBM supercomputer) and AlphaZero (self-training neural-network engine) were not part of these lists, so their estimated Elo values are marked separately. Ratings for humans are FIDE peaks.

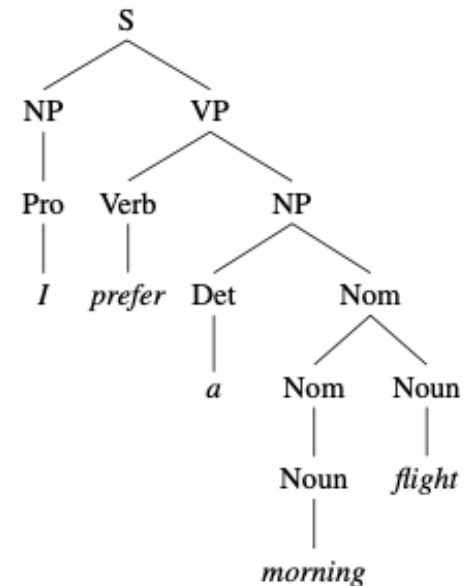
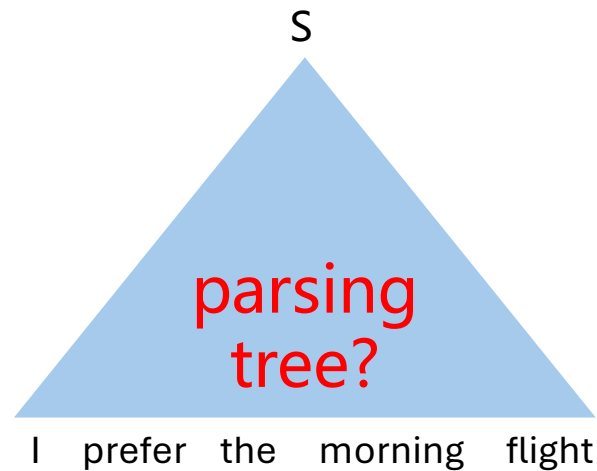
Syntactic parsing

- Goal: given a CFG, assign valid parsing trees to a sentence.
 - Find trees **rooted at S** with **leaves being the words of a sentence**.
 - Reverse the process of generation of sentences using a CFG.

CFG

Grammar Rules	Examples
$S \rightarrow NP VP$	I + want a morning flight
$NP \rightarrow$ Pronoun	I
Proper-Noun	Los Angeles
Det Nominal	a + flight
Nominal $\rightarrow$ Nominal Noun	morning + flight
Noun	flights
$VP \rightarrow$ Verb	do
Verb NP	want + a flight
Verb NP PP	leave + Boston + in the morning
Verb PP	leaving + on Thursday
$PP \rightarrow$ Preposition NP	from + Los Angeles

Noun $\rightarrow$	flights breeze trip morning
Verb $\rightarrow$	is prefer like need want fly
Adjective $\rightarrow$	cheapest non-stop first latest
other direct	
Pronoun $\rightarrow$	me I you it
Proper-Noun $\rightarrow$	Alaska Baltimore Los Angeles
Chicago United American	
Determiner $\rightarrow$	the a an this these that
Preposition $\rightarrow$	from to on near
Conjunction $\rightarrow$	and or but



generation of a
sentence from S

Syntactic parsing: search

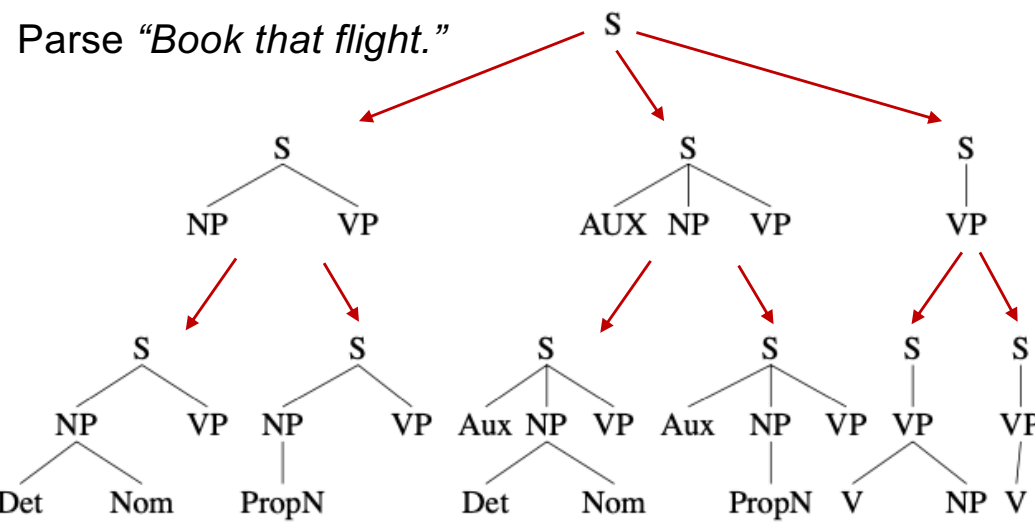
• Top-down search

- Start from the partial parsing tree with *root* = *S* only.
- Expand a non-terminal on an unfinished parsing tree, using a rule in the CFG.
- Pruning: when there is no hope of matching the words in the input sentence.
- Stop with a match of the words in the input sentence.

Pointing from one parsing tree to another parsing tree between two levels in the search tree.

$S \rightarrow NP VP$	$Det \rightarrow that \mid this \mid a$
$S \rightarrow Aux NP VP$	$Noun \rightarrow book \mid flight \mid meal \mid money$
$S \rightarrow VP$	$Verb \rightarrow book \mid include \mid prefer$
$NP \rightarrow Det Nominal$	$Aux \rightarrow does$
$Nominal \rightarrow Noun$	
$Nominal \rightarrow Noun Nominal$	$Prep \rightarrow from \mid to \mid on$
$NP \rightarrow Proper-Noun$	$Proper-Noun \rightarrow Houston \mid TWA$
$VP \rightarrow Verb$	
$VP \rightarrow Verb NP$	$Nominal \rightarrow Nominal PP$

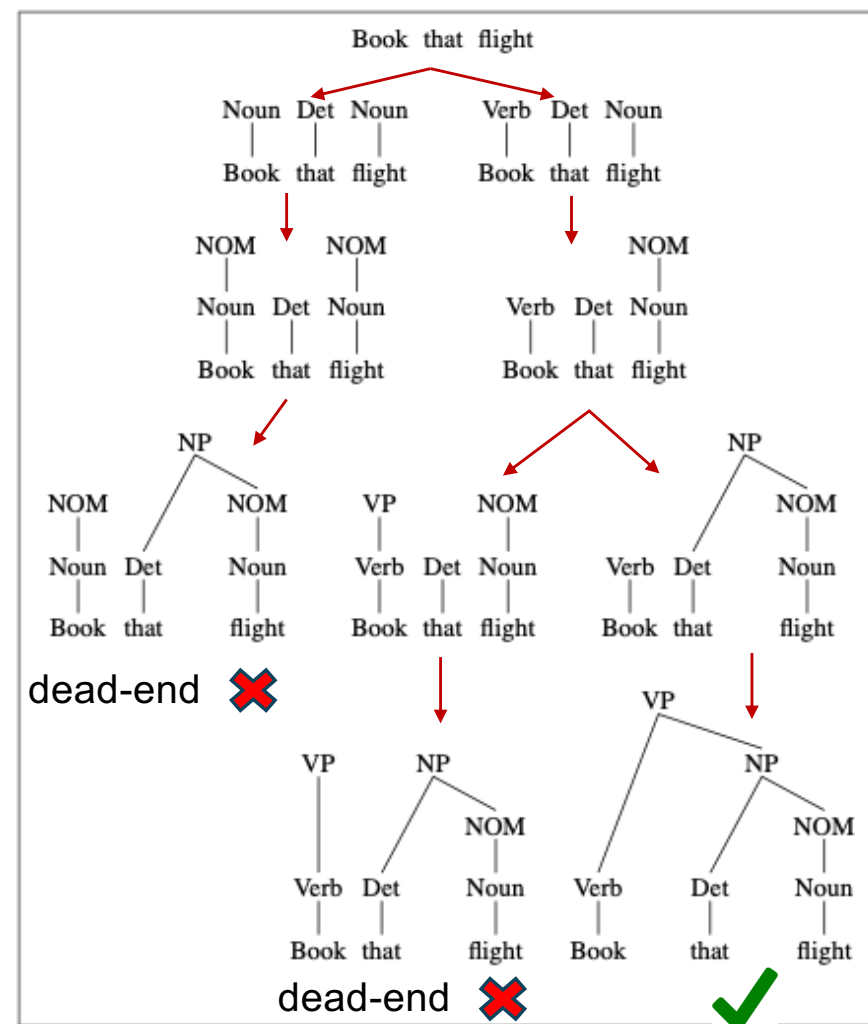
dead-end ❌



Syntactic parsing: search

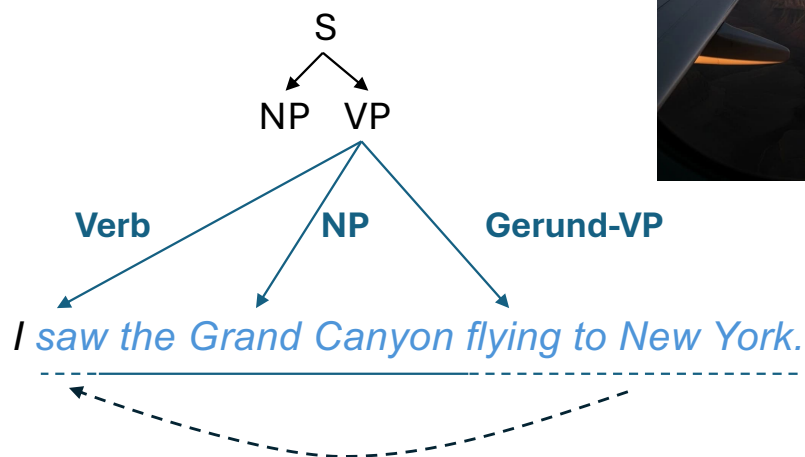
- Bottom-up search
 - start from the leaves (bottom) = input sentence.
 - create a non-terminal from existing symbols, using a rule in the given CFG.
 - Pruning: when can't find a match of any right-hand-side match
 - Stop when arriving at root = S.

$S \rightarrow NP VP$	$Det \rightarrow that \mid this \mid a$
$S \rightarrow Aux NP VP$	$Noun \rightarrow book \mid flight \mid meal \mid money$
$S \rightarrow VP$	$Verb \rightarrow book \mid include \mid prefer$
$NP \rightarrow Det Nominal$	$Aux \rightarrow does$
$Nominal \rightarrow Noun$	
$Nominal \rightarrow Noun Nominal$	$Prep \rightarrow from \mid to \mid on$
$NP \rightarrow Proper-Noun$	$Proper-Noun \rightarrow Houston \mid TWA$
$VP \rightarrow Verb$	
$VP \rightarrow Verb NP$	$Nominal \rightarrow Nominal PP$



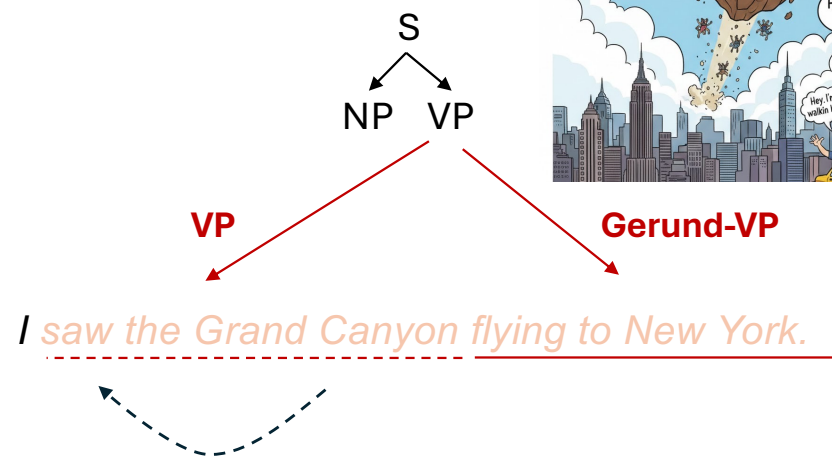
Syntactic parsing: aimbiguity

- Attachment ambiguity



The Gerund-VP is attached to the central Verb *saw*, meaning that, the action “saw” happened as flying to New York.

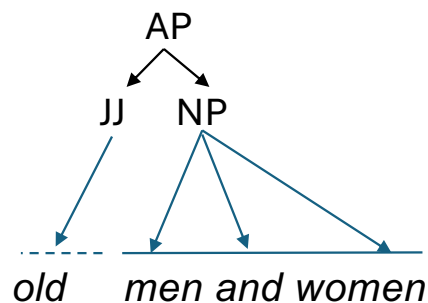
Image sources: <https://www.scenic.com/> and Gemini



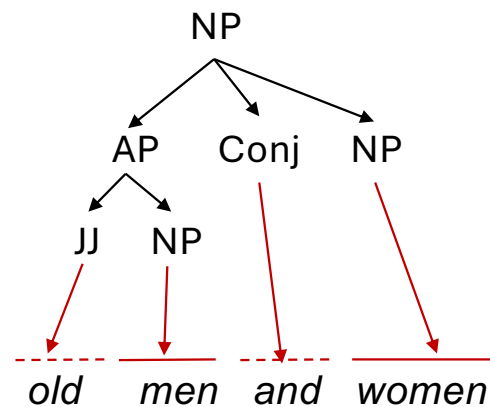
In the lower **VP**, “the Grand Canyon” is attached to the central verb “saw”; in the upper VP, “flying to New York” is attached to “the Grand Canyon”, meaning that the Canyon is flying to New York, and that’s funny.

| Syntactic parsing: aimbiguity

- Coordination ambiguity



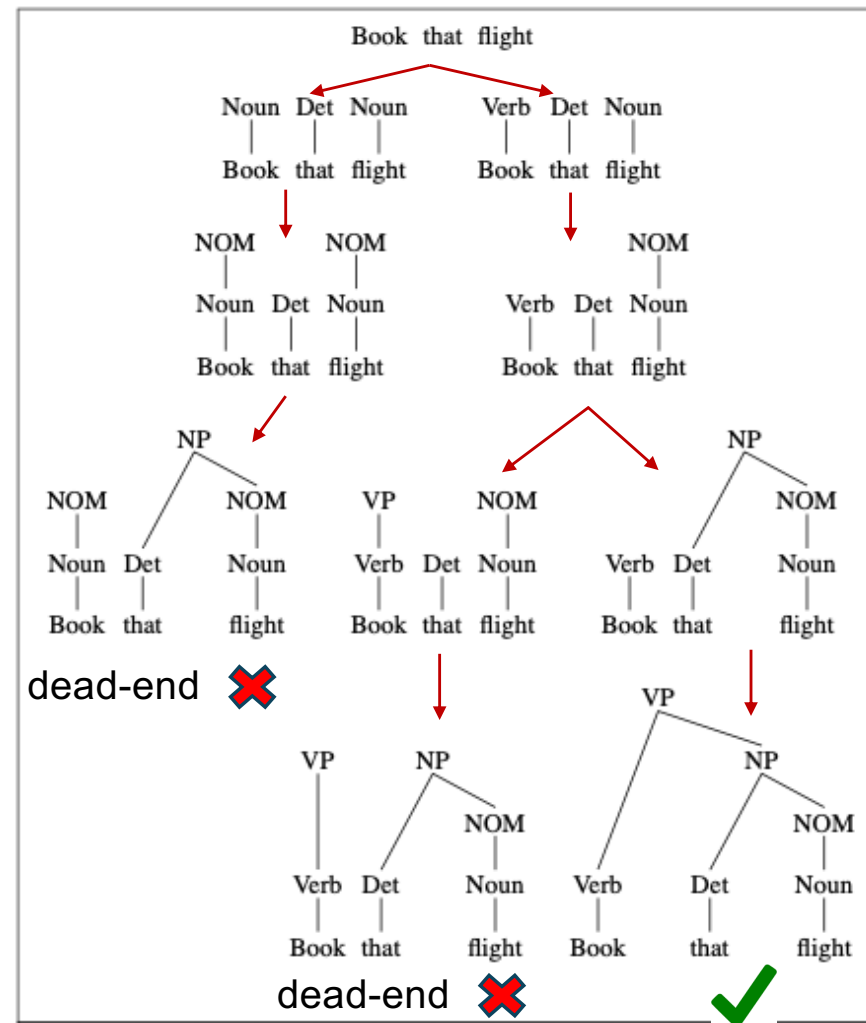
The “*men and women*” is a NP so that the JJ “*old*” is modifying the entire NP including both men and women.



“*old*” and “*men*” are first combined in an AP (Adjective Phrase), which is then combined with the Conjunction “*and*” and the NP “*women*” to form a larger NP. Therefore, old is not modifying “*women*”.

Syntactic parsing: repeated subproblems

- During the search, certain incomplete parsing trees are repeatedly constructed, even some of the trees lead to dead-ends.
 - Det->that, Noun -> flight.
 - NOM->Noun->flight
 - NP -> Det NOM covering "that flight".
- In the bottom-up search, one cannot prune some partial trees early, as only deep down the search tree can the algorithm figure out that it leads to a dead-end.
- Similar things happen to top-down search.



CYK parsing algorithm

- It is a dynamic programming (DP) algorithm.
 - Overlapping sub-problems: can re-use solutions to sub-problems;
 - Subproblems = parsing a sub-part of a sentence.
 - Don't parse the same sub-part multiple times!
 - Optimal structures: optimal solutions => optimal sub-problem solutions.
 - Successful parse of a constituent => successful parse of its sub-parts.
 - A wrong parse of a sub-part won't give a correct parse of a bigger part.
- Reduce the cost of search by caching the solution to subproblems.
 - don't repeatedly parsing "*a flight*".
 - keep the successful parses only.

CYK parsing: CNF CFG

- Chomsky normal form (CNF)
 - Allows a simple data structure for the DP algorithm called “CKY”.
- Restrict CFG to a CNF form:
 - There can be at most two symbols on the right hand side of any rule;
 - If there are two, both should be non-terminals (e.g. $A \rightarrow BC$);
 - If there is one, it should be a terminal (e.g. $A \rightarrow a$).
- There is no loss of expressiveness by the following conversions:
 - mixed production $A \rightarrow Bc \Rightarrow (A \rightarrow BC \text{ and } C \rightarrow c)$
 - $A \rightarrow B \Rightarrow$ eliminate B by changing $A \rightarrow B$ to $A \rightarrow \gamma$ for any $B \rightarrow \gamma$.
 - long right hand side $A \rightarrow BCD \Rightarrow (A \rightarrow XD \text{ and } X \rightarrow BC)$;
 - can handle more than three symbols by recursively doing this conversion.

CYK parsing: CNF CFG

Conversion algorithm

1. handle mixed RHS
2. handle unit productions
3. handle long RHS

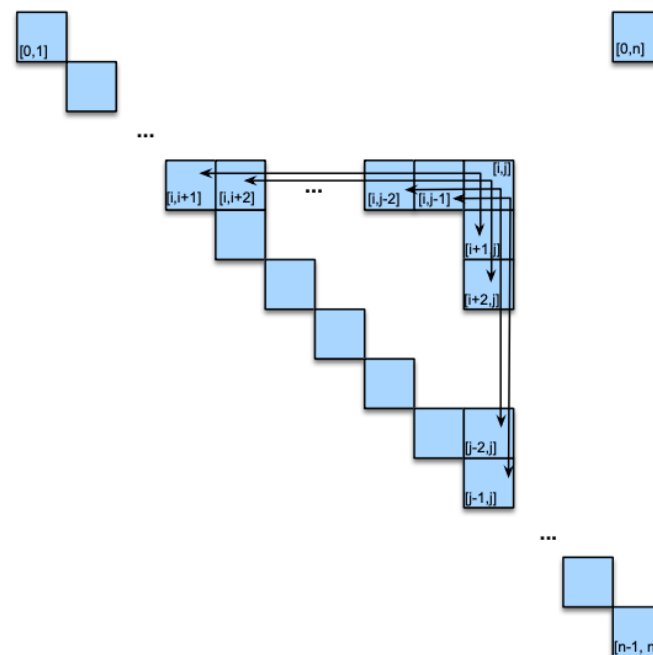
$\mathcal{L}_1$ Grammar	$\mathcal{L}_1$ in CNF
$S \rightarrow NP VP$	$S \rightarrow NP VP$
$S \rightarrow Aux NP VP$	$S \rightarrow X1 VP$
$S \rightarrow VP$	$X1 \rightarrow Aux NP$
	$S \rightarrow book \mid include \mid prefer$
	$S \rightarrow Verb NP$
	$S \rightarrow X2 PP$
	$S \rightarrow Verb PP$
	$S \rightarrow VP PP$
$NP \rightarrow Pronoun$	$NP \rightarrow I \mid she \mid me$
$NP \rightarrow Proper-Noun$	$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det Nominal$	$NP \rightarrow Det Nominal$
$Nominal \rightarrow Noun$	$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal Noun$	$Nominal \rightarrow Nominal Noun$
$Nominal \rightarrow Nominal PP$	$Nominal \rightarrow Nominal PP$
$VP \rightarrow Verb$	$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb NP$	$VP \rightarrow Verb NP$
$VP \rightarrow Verb NP PP$	$VP \rightarrow X2 PP$
	$X2 \rightarrow Verb NP$
$VP \rightarrow Verb PP$	$VP \rightarrow Verb PP$
$VP \rightarrow VP PP$	$VP \rightarrow VP PP$
$PP \rightarrow Preposition NP$	$PP \rightarrow Preposition NP$

CYK parsing: algorithm

- Use a matrix of size $(n+1) \times (n+1)$ if the sentence has n words.

0 Book 1 the 2 flight 3 through 4 Houston 5

- The cell (i, j) contains **all possible** non-terminals that can generate the constituent ranging i to j .
- Due to CNF, each constituent (i, j) with $j > i + 1$ can be broken down into two sub-constituents:
 - $(i, j) = (i, k) + (k, j)$ (e.g., $(1, 3) = (1, 2) + (2, 3)$)
 - $j > i + 1$ and $0 \leq i < k < j \leq n$
- (i, k) and (k, j) represent subproblems that have been solved and stored in cells to the left of and below the cell (i, j) .



CYK parsing: algorithm

function CYK-PARSE(*words*, *grammar*) **returns** *table*

for $j \leftarrow$ from 1 to LENGTH(*words*) **do**

for all $\{A \mid A \rightarrow \text{words}[j] \in \text{grammar}\}$

$\text{table}[j-1, j] \leftarrow \text{table}[j-1, j] \cup A$ // POS tag the j -th word

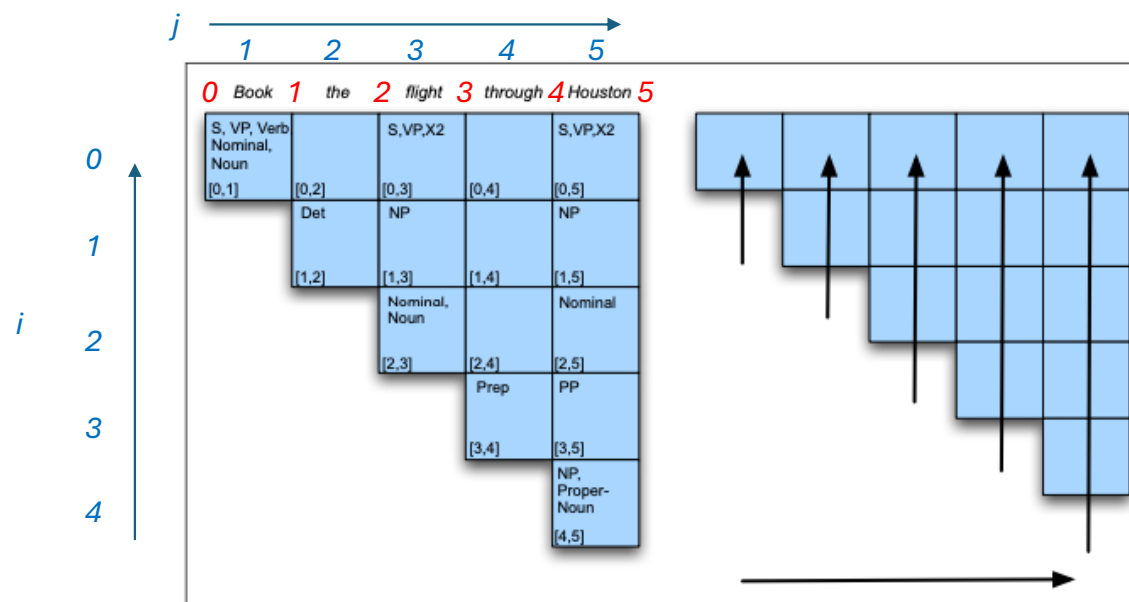
for $i \leftarrow$ from $j-2$ down to 0 **do**

for $k \leftarrow i+1$ to $j-1$ **do** // try all possible breaking points

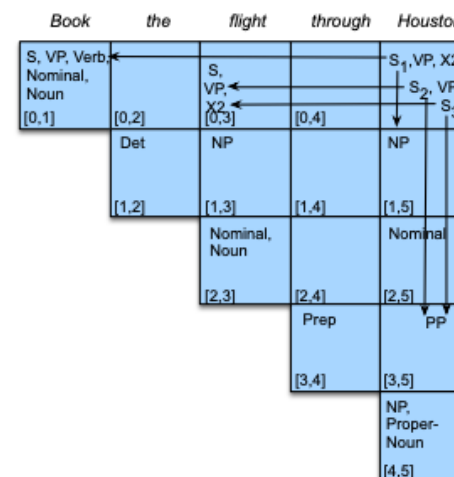
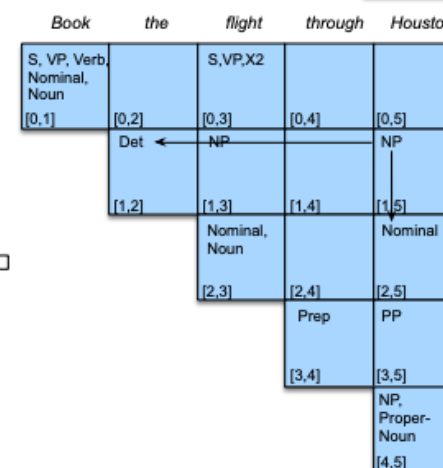
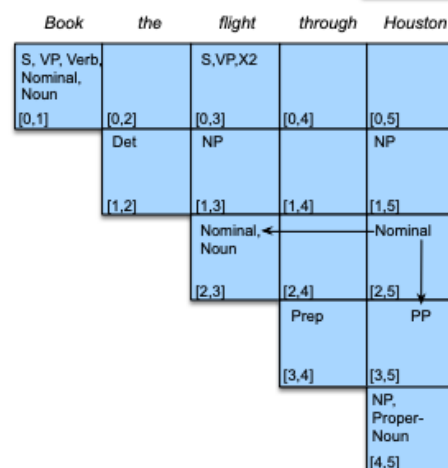
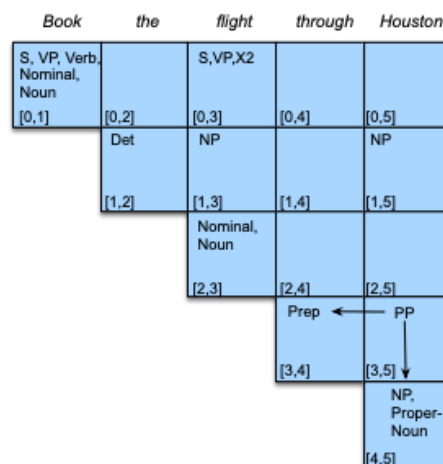
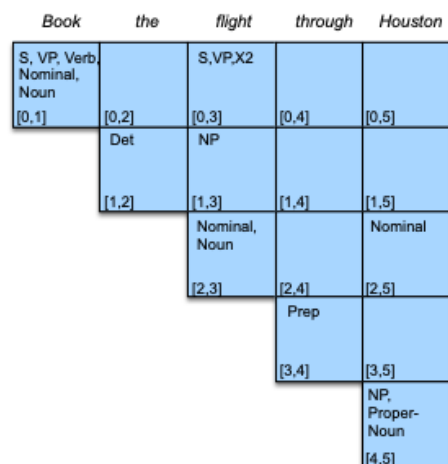
for all $\{A \mid A \rightarrow BC \in \text{grammar} \text{ and } B \in \text{table}[i, k] \text{ and } C \in \text{table}[k, j]\}$

$\text{table}[i, j] \leftarrow \text{table}[i, j] \cup A$ // combine two sub-trees

// try to combine *B* and *C* into a larger part that covers *i* to *j*, breaking at *k*



CYK parsing: examples



$\mathcal{L}_1$ Grammar	$\mathcal{L}_1$ in CNF
$S \rightarrow NP VP$	$S \rightarrow NP VP$
$S \rightarrow Aux NP VP$	$S \rightarrow X1 VP$
	$X1 \rightarrow Aux NP$
$S \rightarrow VP$	$S \rightarrow book \mid include \mid prefer$
	$S \rightarrow Verb NP$
	$S \rightarrow X2 PP$
	$S \rightarrow Verb PP$
	$S \rightarrow VP PP$
$NP \rightarrow Pronoun$	$NP \rightarrow I \mid she \mid me$
$NP \rightarrow Proper-Noun$	$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det Nominal$	$NP \rightarrow Det Nominal$
$Nominal \rightarrow Noun$	$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal Noun$	$Nominal \rightarrow Nominal Noun$
$Nominal \rightarrow Nominal PP$	$Nominal \rightarrow Nominal PP$
$VP \rightarrow Verb$	$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb NP$	$VP \rightarrow Verb NP$
$VP \rightarrow Verb NP PP$	$VP \rightarrow X2 PP$
	$X2 \rightarrow Verb NP$
$VP \rightarrow Verb PP$	$VP \rightarrow Verb PP$
$VP \rightarrow VP PP$	$VP \rightarrow VP PP$
$PP \rightarrow Preposition NP$	$PP \rightarrow Preposition NP$

Since the cell [0,5] contains starting symbols, so the parsing has found multiple complete parsing trees.

| Extra readings

- For CFG, see Chap 18.2 of SLP3.
- For more examples of parsing ambiguity, see Chap 18.5 of SLP3.
- For CNF and CYK, see Chap 18.6 of SLP3.
- Bitter lesson: see <http://www.incompleteideas.net/IncIdeas/BitterLesson.html>

| Demon

- First generate a CFG in CNF with a small number of non-terminals and terminals.
- Generate sentences from the CFG through random sampling.
- Implement the CYK algorithm to parse the generated sentences.
- Visualize some of the parsing trees.

| Conclusion

- Syntax is an important aspects of a language, concerning how words are organized.
- Shallow vs. deep syntax.
- Search vs. DP for syntactic parsing.

Quiz

1. Pen and paper; no compute or cellphone allowed.
2. Turn in your answer sheet when you leave the classroom.

- Q1 (T/F): the above CFG is in CNF.
- A1: T
- Q2 (short answer): write down all the needed derivations to generate the sentence "the big dog eats a fish".
- A2: $S \rightarrow NP VP$, $NP \rightarrow Det NP$, $NP \rightarrow Adj N$, $VP \rightarrow V NP$, $NP \rightarrow Det NP$, $Det \rightarrow$ "the", $Det \rightarrow$ "a", $Adj \rightarrow$ "big", $N \rightarrow$ "dog", $N \rightarrow$ "fish"
- Q3 (short answer): when running the CYK algorithm on the above sentence, what's in the cell (0,2) when CYK finishes?
- A3: nothing, since "the big" in the range of (0,2) cannot be derived from the above CFG.

$S \rightarrow NP VP$
 $VP \rightarrow V NP$
 $NP \rightarrow Det N$
 $NP \rightarrow Adj N$
 $NP \rightarrow Det NP$
 $Det \rightarrow$ "the" | "a"
 $N \rightarrow$ "cat" | "dog" | "fish"
 $V \rightarrow$ "eats" | "sees" | "likes"
 $Adj \rightarrow$ "big" | "small"